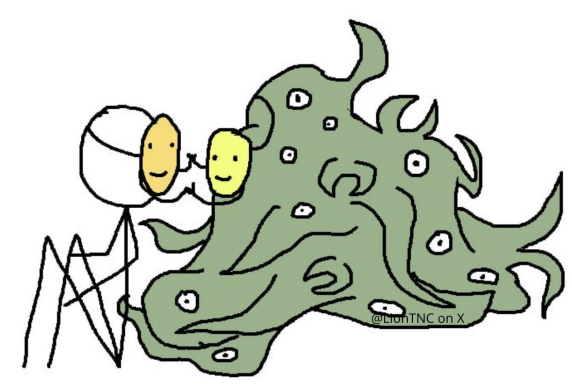


Enhancing Tech Creation through New Multimodal Context



JETBRAINS Research



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XR interaction design space*

How the human-computer interaction during tech creation can be enhanced with the progress in multimodal AI-XR tools?

Our RQs:

- What are the needs and recurring pain points of tech creators that can be addressed with multimodal AI-XR tools
- What types of multimodal interaction techniques are the most promising in addressing these needs and what the AI-XR tooling possibilities for tech creation could look like
- What are the main constraints for building multimodal AI-XR tools for tech creation today

Approach: structured interviews with AI-XR academic and industry experts. First pull of experts was identified as people who delivered keynote speeches or received significant awards at IEEE VR, CHI, VRST, and ISMAR conferences within the last five years.

Pre-registration materials for the XR design space study



*Design space is used in a metaphorical sense (Heape 2007), as the space of design possibilities and alternatives for developing new tools (Luck & Aylett, 2000)

Engineering feasibility

What are technological challenges and their mitigations when building XR-augmented tools for human-AI interaction in tech creation?

(or when do we “hit the wall”)

Our RQs:

- Are current IDEs compatible with common XR platforms?
- What are the technical limitations of popular SDKs (e.g., Apple Vision Pro SDKs, RealityKit)
- What is the most promising technological stack to pursue?

We are looking for engineers to join our efforts. Message @ilia.zaharov if you are interested!

First prototype - XR eye-tracking as a multi-modal input

How do developers experience and perceive XR-based IDEs features compared to traditional ones?

Our RQs:

- Can we use eye-tracking data to make interaction with AI models more efficient, precise and comfortable?
- Can we use eye-tracking data to increase accuracy of next edit prediction models? context understanding? any other types of existing scenarios?

Approach: Conduct comparative user studies evaluating raw XR environments, interactive XR prototypes, and traditional coding setups. Collect both qualitative feedback (user experiences, perceptions) and quantitative data (productivity metrics, cognitive load). The prototype ideas will be extracted from XR design space research

XR we can work with



PICO 4 Enterprise

Eye-tracking data - the positions and orientations of both eyes, rich data about gaze direction, PICO API

Potential connecting to IDE:

- PICO Connect + Virtual Desktop
- Or - browser-based IDEs



Varjo XR-3*

Eye-tracking data -high-fidelity raw data directly from manufacturer

Potential connecting to IDE:

- to be established

*This headset is heavy, connected to cables and is closer to lab settings



Bigscreen Beyond 2

Eye-tracking - gaze position, gaze movements, OpenXR standard

Potential connecting to IDE

- Through SteamVR (no browser-based IDEs)

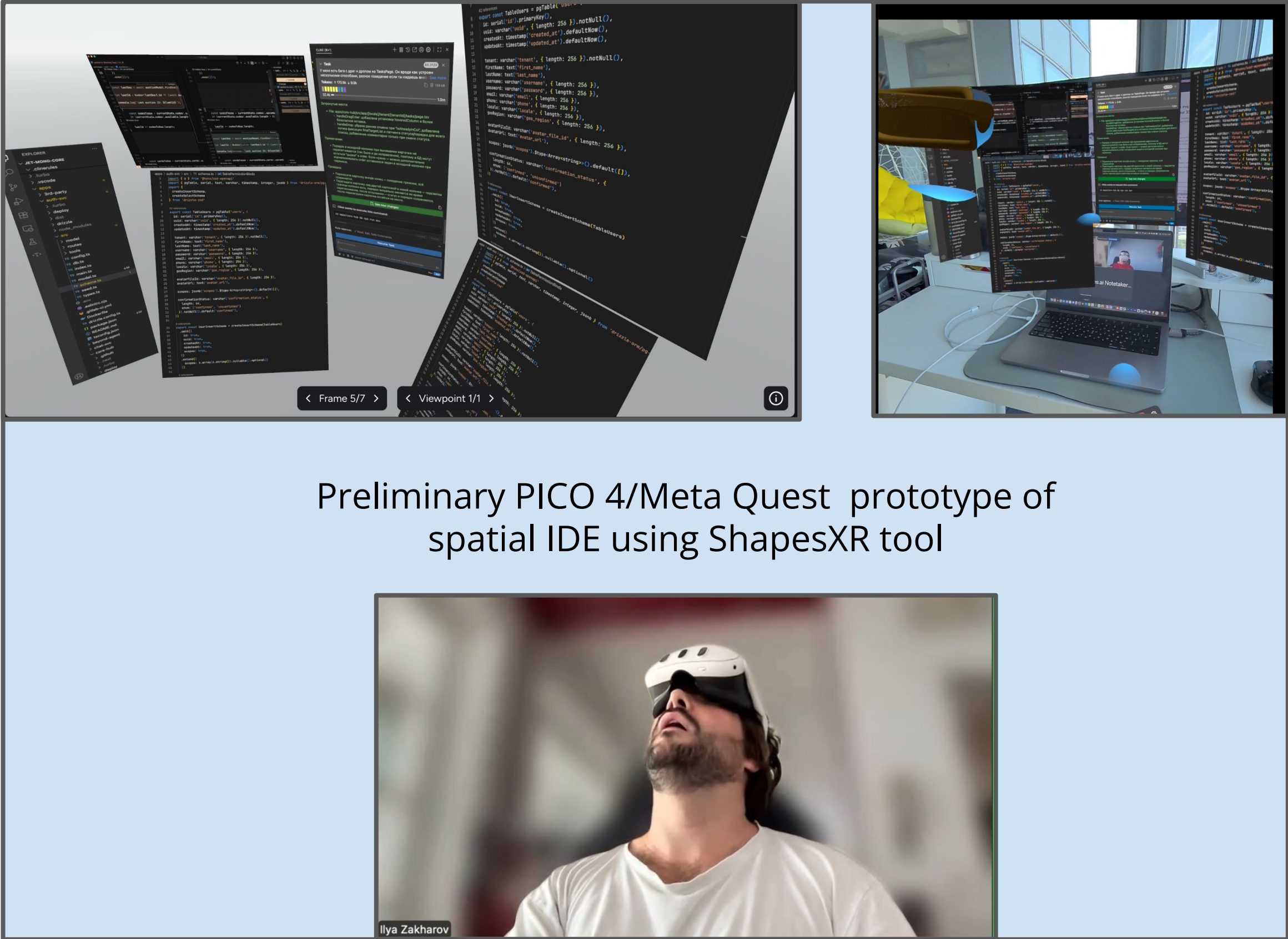


Apple Vision Pro

Eye-tracking data - gaze focus on “interactive objects” (defined through visionOS)

Potential connecting to IDE

- Vision Pro as extended monitor for Mac (also can be stream to Windows)
- VisionCode for visionOS



Preliminary PICO 4/Meta Quest prototype of spatial IDE using ShapesXR tool

Future plans

Expanding Prototype Functionality:

- More prototypes based on the design space and feasibility report.

Extending XR:

- Not only virtual, but augmented reality (AR)

Advanced Evaluation and Human Factors:

- creating benchmarks
- prolonged using of headsets

Broadening the multimodal scope:

- More inputs (wearable data, external sensors, mouse tracking, etc.)

Follow the project in YT:



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